

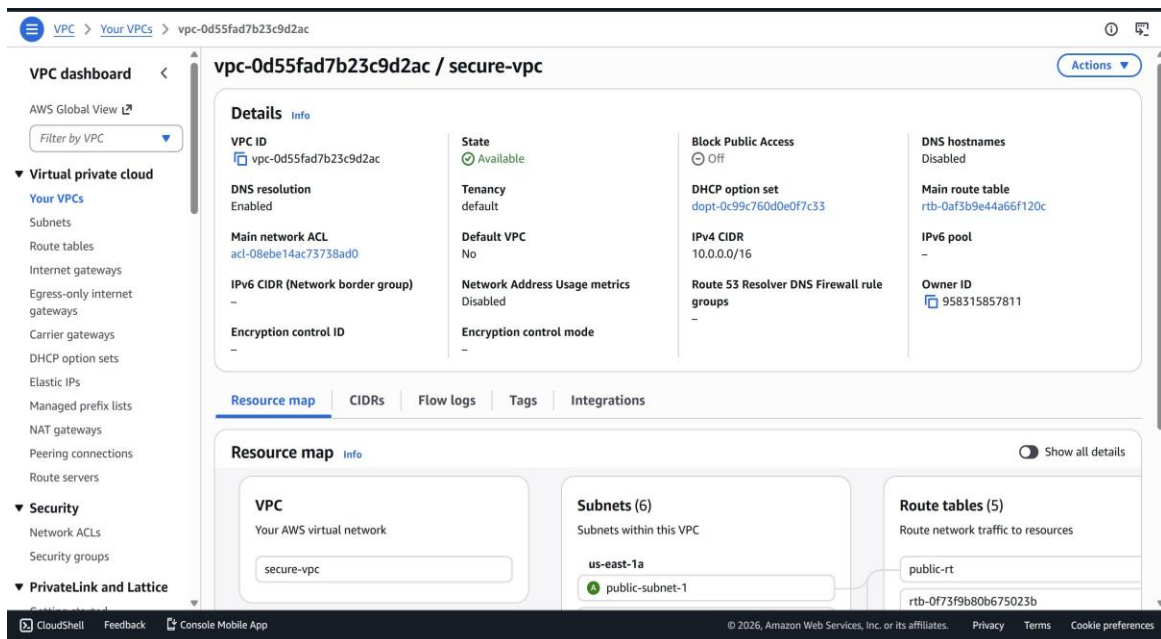
AWS SECURE CLOUD ARCHITECTURE

Evidence Screenshots — Infrastructure Verification

These screenshots provide visual verification of the AWS infrastructure built for the Defense-in-Depth security architecture project. Each screenshot corresponds to a specific component documented in the main architecture document.

1. VPC Configuration

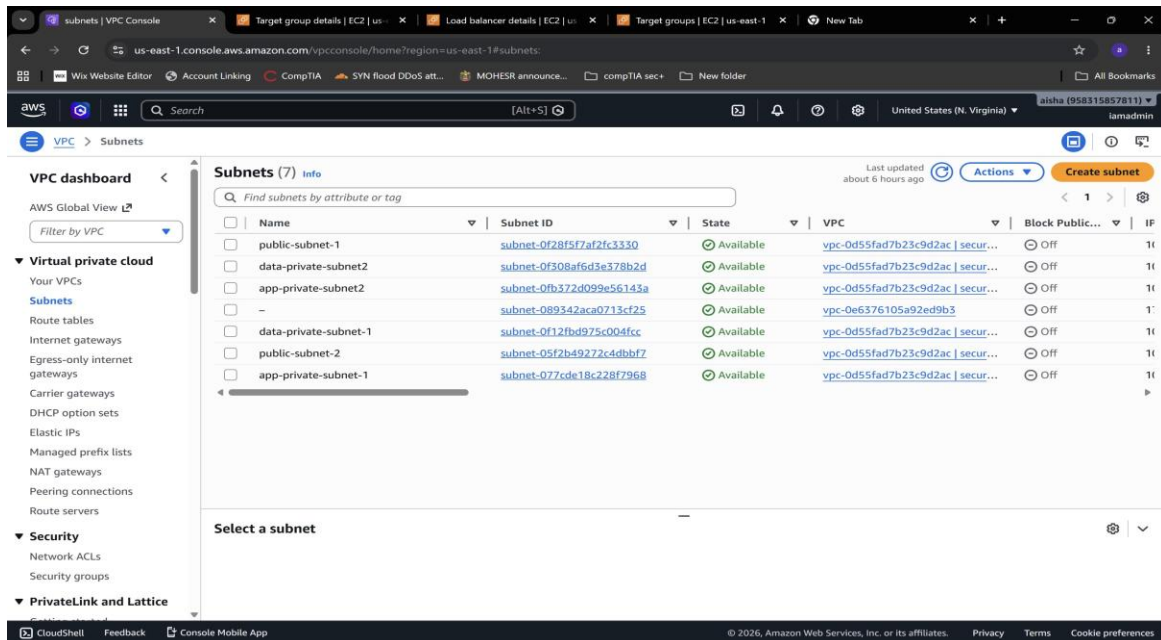
The secure-vpc was created with CIDR 10.0.0.0/16, providing an isolated network environment for all architecture components.



Screenshot: secure-vpc — VPC ID vpc-0d55fad7b23c9d2ac, CIDR 10.0.0.0/16

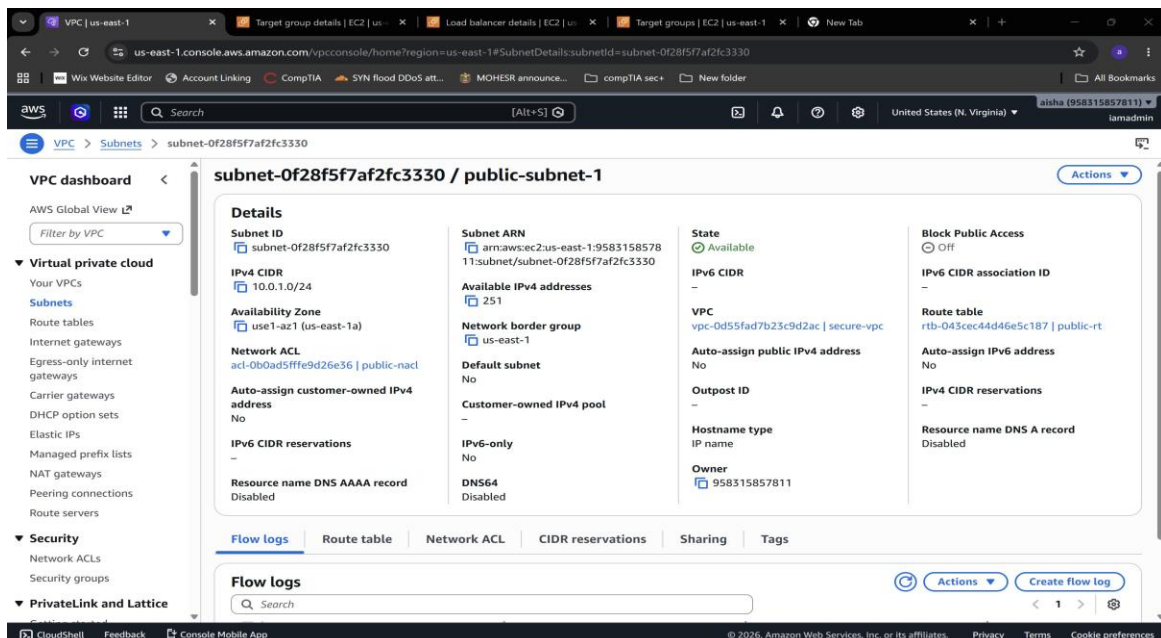
2. Subnet Configuration

Six subnets were created across two Availability Zones — two public, two application private, and two data private — implementing the 3-tier network segmentation.



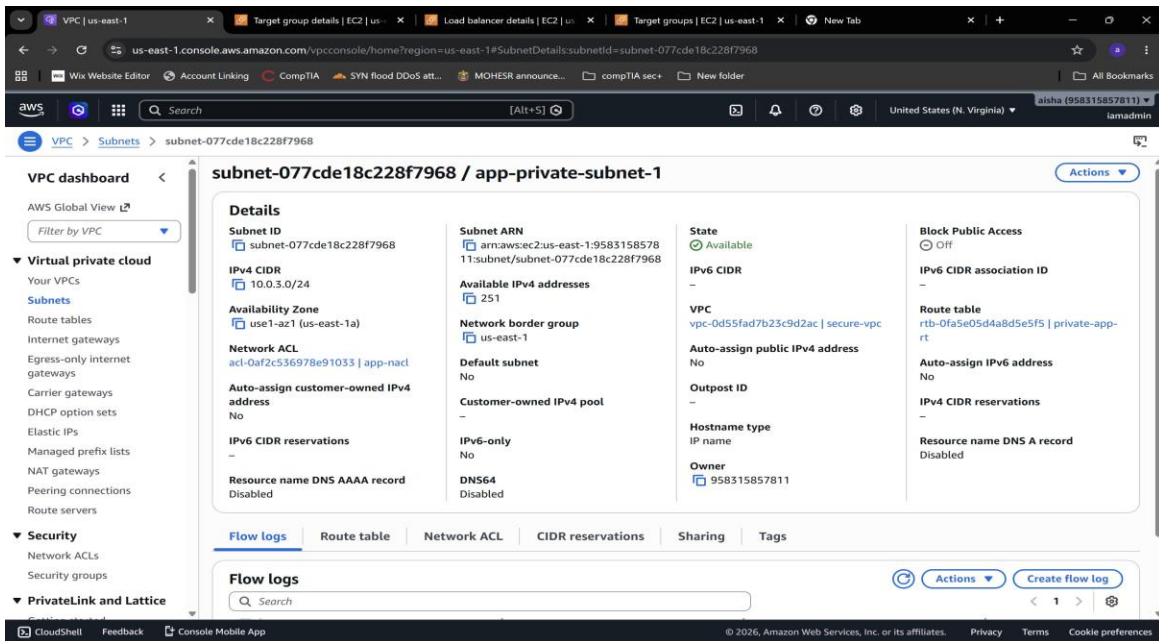
Screenshot: All 6 subnets — public, app-private, and data-private tiers across AZ-a and AZ-b

2.1 Public Subnet



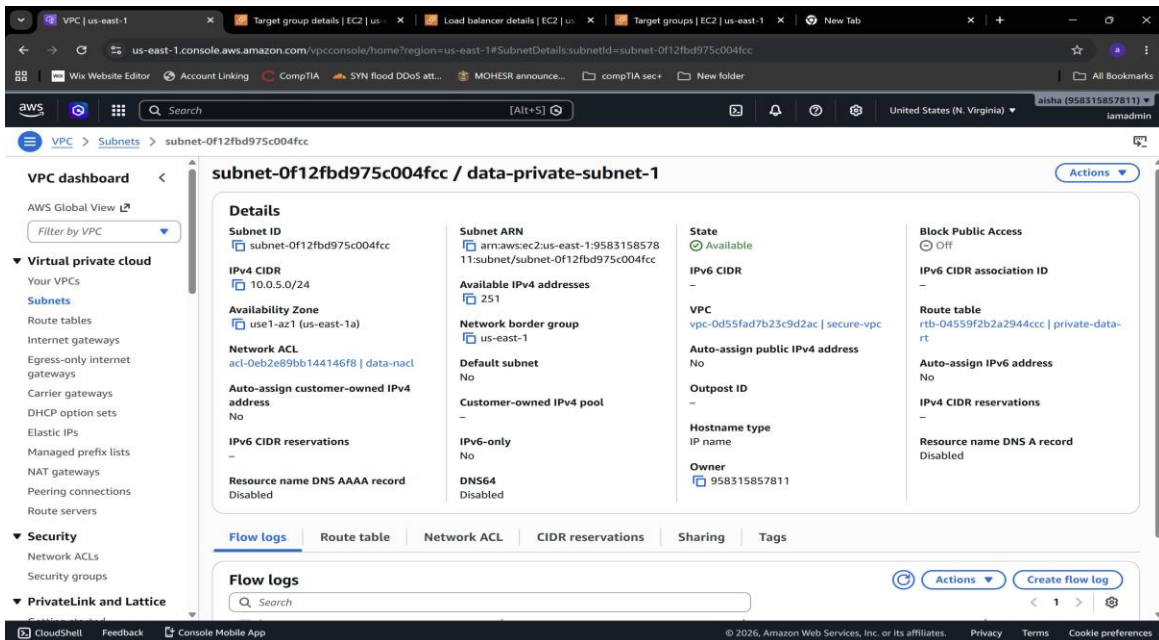
Screenshot: public-subnet-1 — CIDR 10.0.1.0/24, AZ us-east-1a, associated with public-rt

2.2 App Private Subnet



Screenshot: `app-private-subnet-1` — CIDR `10.0.3.0/24`, AZ `us-east-1a`, no public IP assignment

2.3 Data Private Subnet



Screenshot: `data-private-subnet-1` — CIDR `10.0.5.0/24`, AZ `us-east-1a`, associated with `data-nac1`

3. Route Tables

Three route tables implement tiered internet access — public subnets route to IGW, app subnets route to NAT Gateway, data subnets have local-only routing.

3.1 Public Route Table

The screenshot shows the AWS VPC console interface. On the left, the 'VPC dashboard' sidebar is visible with categories like Virtual private cloud, Security, and PrivateLink and Lattice. The main area displays 'Route tables (1/6)'. A table lists several route tables, with 'public-rt' (ID: rtb-043cec44d46e5c187) selected. Below the table, the 'Routes' tab for 'public-rt' is active, showing two routes:

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0264f98a2479bedc0	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Screenshot: public-rt — routes: 0.0.0.0/0 → IGW, 10.0.0.0/16 → local

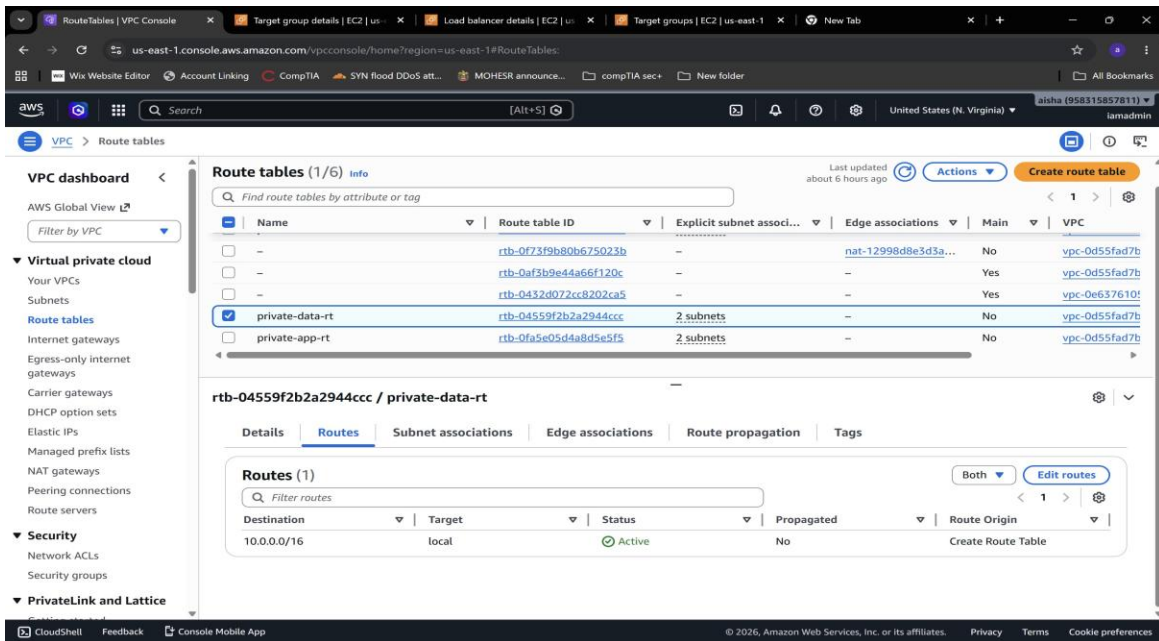
3.2 App Private Route Table

The screenshot shows the AWS VPC console interface. On the left, the 'VPC dashboard' sidebar is visible. The main area displays 'Route tables (1/6)'. A table lists several route tables, with 'private-app-rt' (ID: rtb-0fa5e05d4a8d5e5f5) selected. Below the table, the 'Routes' tab for 'private-app-rt' is active, showing two routes:

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-12998d8e3d3a26f45	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Screenshot: private-app-rt — routes: 0.0.0.0/0 → NAT Gateway, 10.0.0.0/16 → local

3.3 Data Private Route Table

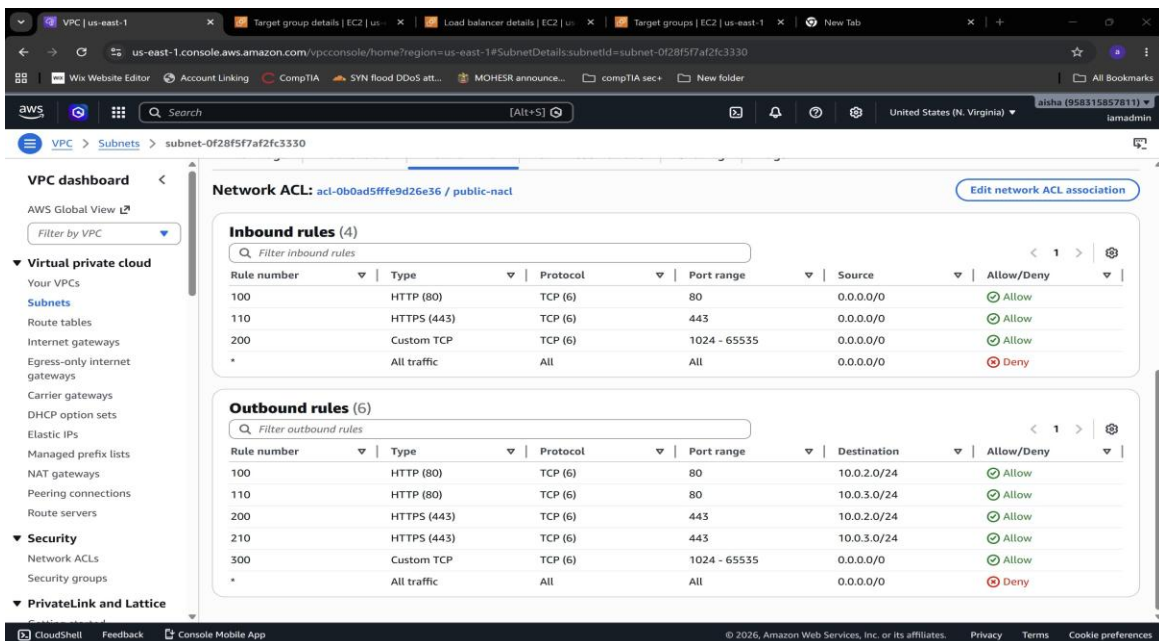


Screenshot: private-data-rt — routes: 10.0.0.0/16 → local only (no internet access)

4. Network Access Control Lists (NACLs)

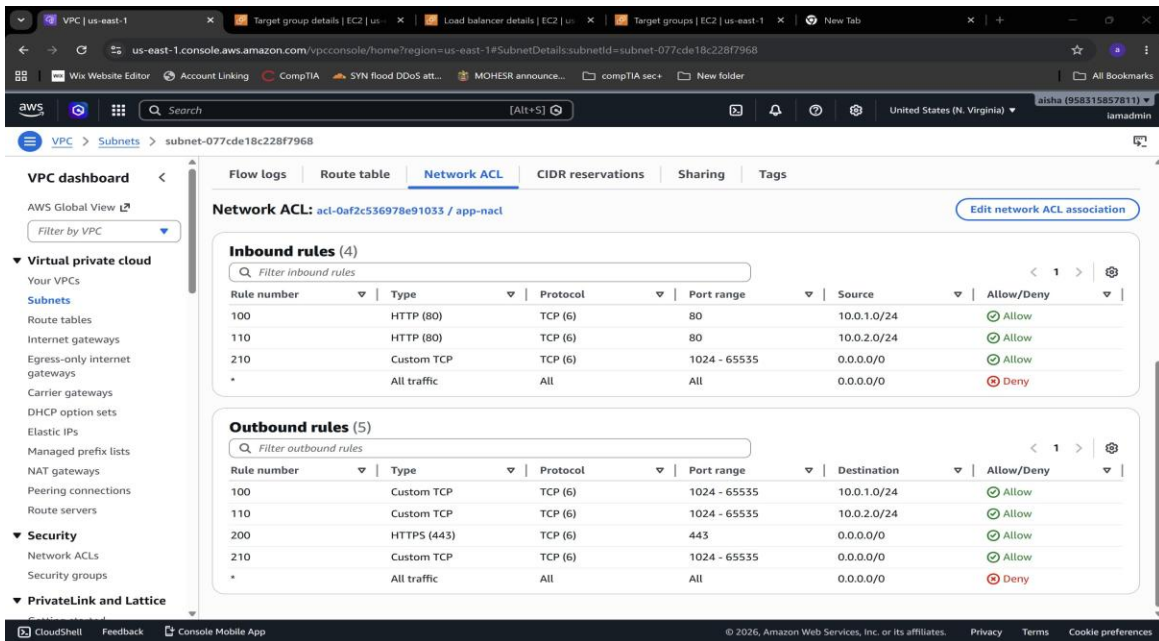
Three NACLs provide stateless subnet-level packet filtering. Each NACL is scoped to allow only the minimum required traffic for its tier.

4.1 Public NACL



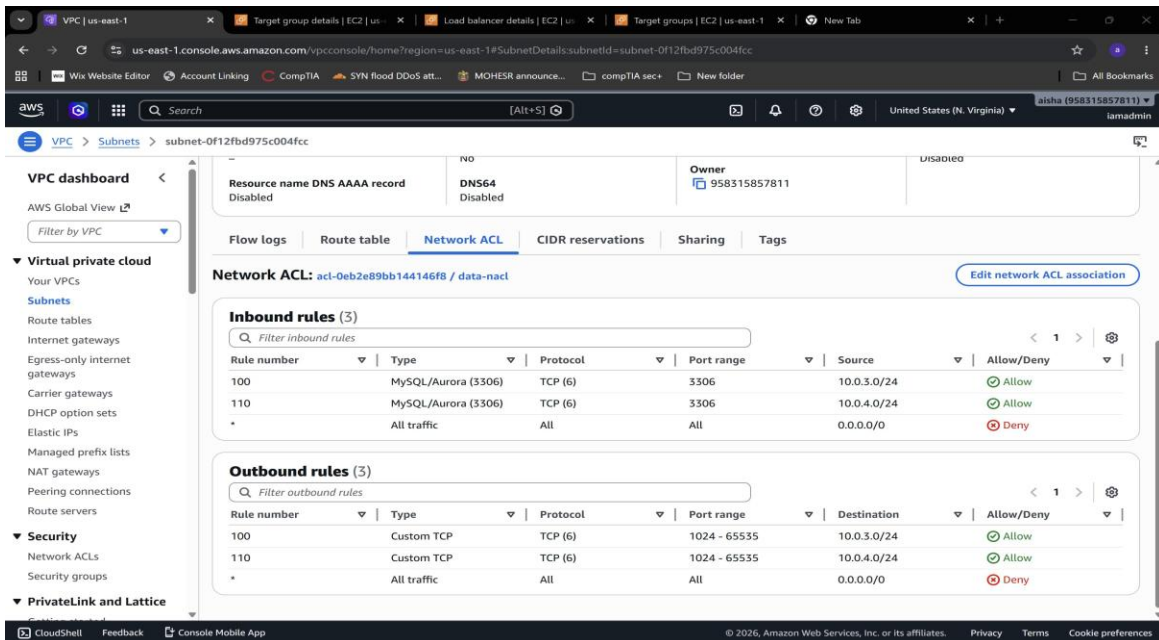
Screenshot: public-nacl — inbound: HTTP/HTTPS/ephemeral from internet; outbound: to app subnets and internet

4.2 App NACL



Screenshot: app-nacl — inbound: port 80 from public subnets only; outbound: responses + HTTPS for SSM/updates

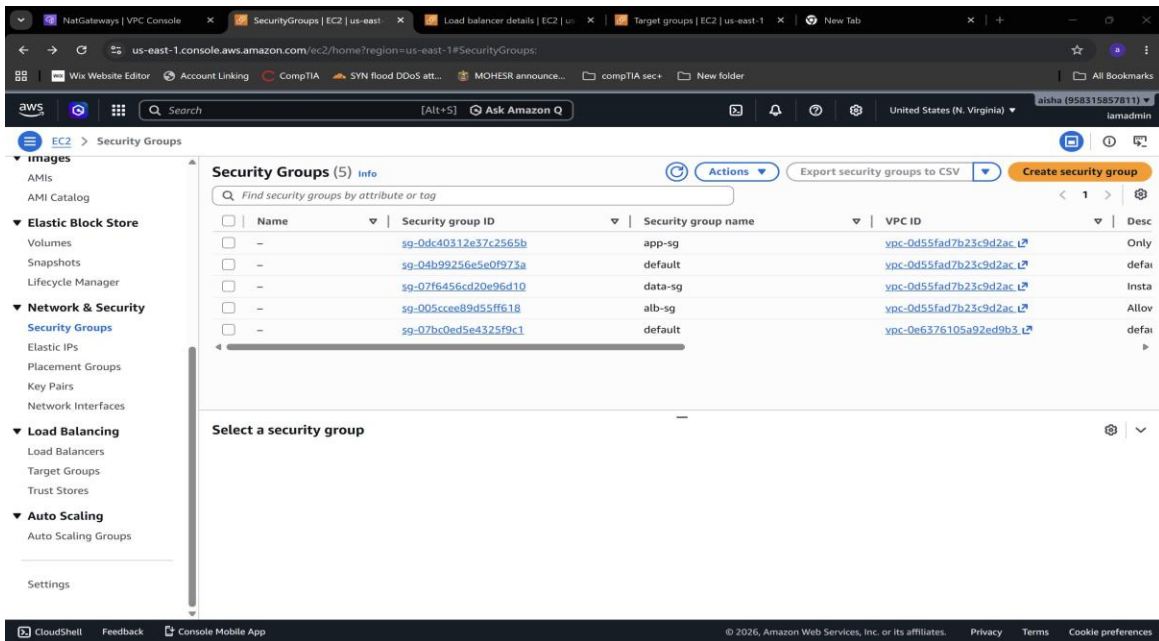
4.3 Data NACL



Screenshot: data-nacl — inbound: port 3306 from app subnets only; outbound: ephemeral ports to app subnets only

5. Security Groups

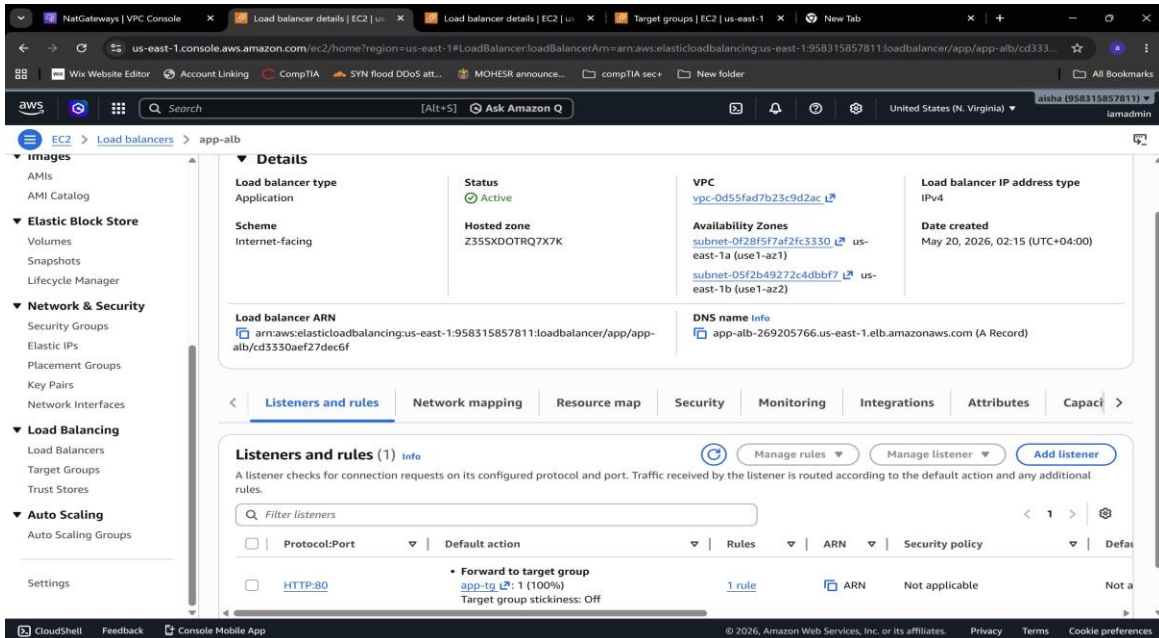
Four security groups implement stateful instance-level access control with SG-to-SG referencing — no IP-based rules used anywhere.



Screenshot: Security Groups — alb-sg, app-sg, data-sg all in secure-vpc

6. Application Load Balancer

The Application Load Balancer is deployed in public subnets across two AZs, configured as internet-facing with HTTP:80 listener forwarding to the app-tg target group.



Screenshot: app-alb — Internet-facing ALB, Active status, deployed across us-east-1a and us-east-1b